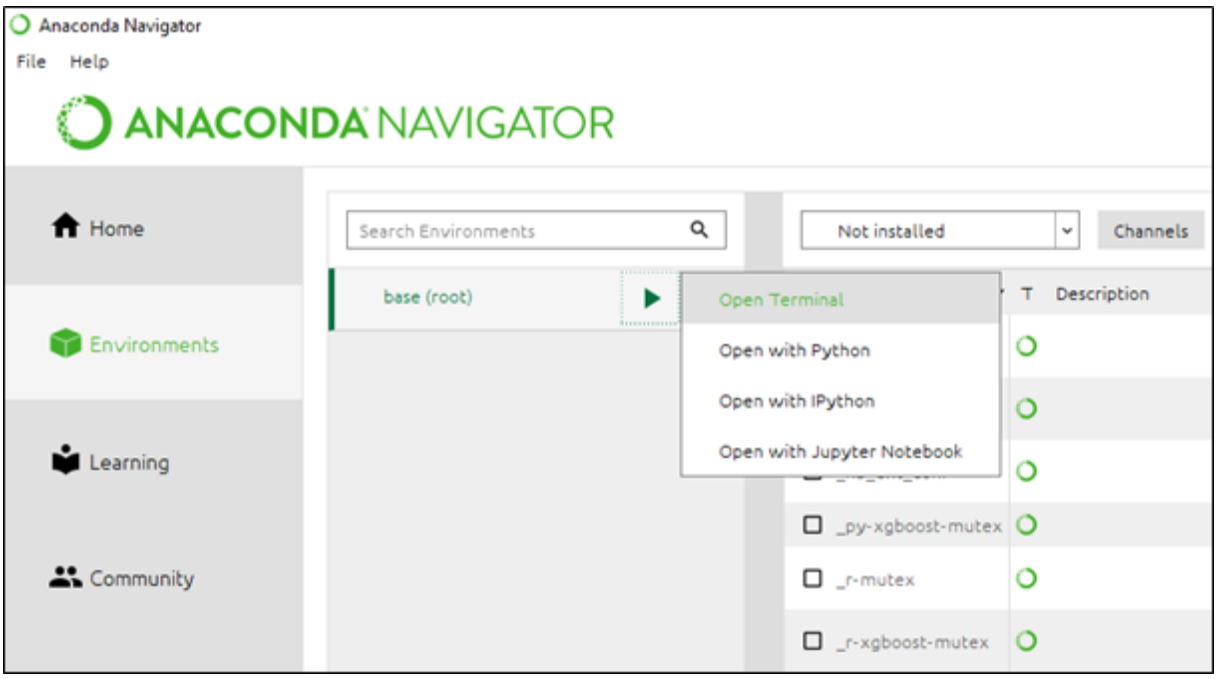
Simulation Assignment

For this assignment, we will be modeling a jewelry business. Make sure that the 'simpy' simulator is installed, by doing the following:

- make sure that the Spyder IDE is closed;
- start anaconda navigator;
- in the tabs to the left, click 'Environments' (see figure below);
- in your base environment, click the 'play' button;
- click 'Open Terminal';
- in the terminal that opens now, type 'pip install simpy' and press enter.

Simpy should now be installed. When the installation is completed, you can close the terminal and start Spyder.



Exercise 1 (30 points)

In this exercise, you will simulate the design and production process of a research and development department of major tire manufacturer. This department receives orders from business engineers of car manufactures to produce tires of a new tire prototype. Your task is to track and optimize the workflow of this process, from the initial designing phase to the final shipping of produced products. This process involves multiple stages, each with its own time constraints and dependencies, which you will model in your simulation.

The production process consists of the following stages:

- Stage 1: Tire design
- Stage 2: Subcomponent Collection
- Stage 3: Tire Assembly
- Stage 4: Tire Curing
- Stage 5: Quality Inspection
- Stage 6: Shipping

Before any order can enter production, the design phase must be completed, which is handled by engineers. Orders can be categorized into two types:

- Type A (Standard Design): Predefined designs.
- Type B (Sports Design): Tailored for specific vehicle models.

Orders arrive at an exponential rate with an average of $\lambda=0.5$/day. This means you'll typically receive an order every two days, but this timing can vary. 90% of these orders will be of type A, and the remaining 10% of type B.

The stages can be further broken down to the following characteristics:

- **Tire Design:** For each order, the design process of tires of type A takes between 2 to 5 days and of type B between 5 and 10 days (both, uniformly distributed).
- **Subcomponent Collection:** 70% of the designs, both for type A and B, can be produced using standard subcomponents that are already in stock. The gathering of these subcomponents takes between 3 and 6 hours (uniformly distributed). For the remaining 30% of the designs, we depend on other production units, that will produce the subcomponents on request. The production of these subcomponents requires a request to produce, taking between 2 to 5 days (uniformly distributed).
- **Tire Assembly:** The assembling of an order takes between 1 to 3 days (uniformly distributed). However, there's a 5% risk that the assembly fails, requiring the regathering of the required subcomponents to retry the assembly. You may assume that the additional regathering of the required components can be done with components that are already inhouse. Furthermore, you may assume that the assembly of an order can fail at most once.
- **Tire Curing:** Once assembled, the order of tires are cured at high temperatures for exactly three days.
- **Quality Inspection:** Each order must pass through a 12-hour inspection. Randomly, 1% of the orders will be selected for additional rigorous testing, which adds an additional 24 hours to the inspection process.
- **Shipping:** Once approved, the order is packed and shipped. This final stage takes 12 hours.

Extend the template file 1.py to create a simulator for this exercise. Assume the production facility operates 24 hours per day, and simulate the process over the time span of 1 year, in hours. Track the orders (per type) at different stages during the process.

For this assignment, you may assume that you have 1 resource available for “Tire Design”, 1 resource for Tire Assembly, 1 for curing, 1 for Quality Inspection and 1 for Shipping. **All resources can solely be used for their specified task.** You can ignore the resources required for the ‘Subcomponent Collection’ phase, as this is managed by the subcomponent production departments.

The solution check will produce approximately the following output:

```
RESULTS
Simulation Duration: 8761.384484244109
Total Arrived Orders A: 163
Total Arrived Orders B: 20
Designed Orders A: 86
Designed Orders B: 9
Assembled Orders A: 85
Assembled Orders B: 9
Cured Orders A: 83
Cured Orders B: 9
Completed Orders A: 83
Completed Orders B: 9

DONE
```

NB: The numbers generated by your solution may differ due to randomness but should be of the same order of magnitude.

Exercise 2 (20 points)

This exercise extends the solution to Exercise 1.

Now, we start recording the processing data and define functions that can perform calculations to provide more insights into the process. You are asked to define the following functions:

1. **compute_resource_utilization**
   This function calculates the fraction of time that resources (design, assembly, curing, quality inspection, packaging) are occupied. The function takes the environment, resource type (as a string: 'design', 'assembly', 'curing', 'quality', or 'shipping'), and the duration of the simulation as inputs. Ensure the simulation is run before calling the function. Return the utilization rate as a string that contains a percentage with 2 significant figures. If an invalid resource type is provided, return: "RESOURCE TYPE DOES NOT EXIST!" Note that you are not allowed to use global variables inside the function. You can use only parameters of the function.

2. **compute_average_cycle_time**
   This function computes the average cycle time of tire orders from arrival to completion of the pack-and-ship operation. The function takes as input the tire type ('A', 'B', or 'Both'). If 'Both' is provided, return the average cycle times for both types combined. If an invalid order type is provided, return: "ORDER TYPE DOES NOT EXIST!" Note that you are not allowed to use global variables inside the function. You can use only parameters of the function.

3. **plot_cycle_times**
   Create a function that visualizes the development of cycle times throughout the simulation as a line plot. The plot should include three lines: one representing orders of type ‘A’, one representing orders of type ‘B’, and one representing orders of both types combined ('Both'). The x-axis should represent the moment of arrival of an order, while the y-axis should show the corresponding processing duration ((hours). Note that you are not allowed to use global variables inside the function. You can use only parameters of the function.

Extend the template file 2.py to create a simulator for this exercise.

NB: No preview is provided for this exercise. Ensure that the output makes logical sense.

Exercise 3 (20 points)

This exercise extends the solution to Exercise 1. **(Note:** Not Exercise 2**)**

We further develop the simulation model of the factory. Update the simulation model from Exercise 1 to include the following:

- **Prioritization:** Prioritize Type B (custom) tire orders over Type A in every operation stage. Preemption is not allowed; operations must finish before starting a new task.
- **Shared resources:** Allow the curing resource to be also used in the assembly stage, and the shipping resource to also work in the inspection stage. Note: the curing resource cannot work at the assembly and the inspection resource cannot work at the shipping stage.
- **Predefined arrivals:** Use a predefined arrival pattern for this assignment, provided in the 'order_arrival_information.json' file. Use the following code to load this data:

  ```
  import json
  with open('order_arrival_information.json') as data:
       arrival_information = json.load(data)
  ```

Extend the template file 3.py to create a simulator for this exercise.

NB: No preview is provided for this exercise. Ensure that the output makes logical sense.

Exercise 4 (20 points)

This exercise extends the solution to Exercise 1. **(Note: Not of Exercise 2)**

In Exercise 1, the factory was unable to complete all incoming orders within the simulation period. The factory now aims to expand its capabilities to achieve an order completion rate of **95%**. You have been hired as a consultant to advise the factory what resources should be hired to achieve this. The costs of additional resources are as follow:

Resource Type:	**Costs:**
Design resource	40.000
Assembly resource	10.000
Curing resource	10.000
Inspection resource	10.000
Shipping resources	10.000

Your task is to write the following function: 'find_best_investment_opportunity(budget)'. This function receives the available investment budget, which should be allocated towards recruiting additional factory resources. Design the function as follow:

1. Get a list of investment options based on the given budget (**use the entire budget**).
2. Iterate over the each option in the investment options.
3. Return the best option and its quality in terms of the completion rate.

The performances of the different options are affected by the randomness in the arrival and processing times. Therefore, you are asked to work with replications (n=5). Make sure that no unnecessary options will be checked. Adhere to the format provided in the template.

Extend the template file 4.py to create a simulator for this exercise.

NB: No preview is provided for this exercise. Ensure that the output makes logical sense.